Execution-Aware Skill Adaptation for Manipulation: Benchmark Pivot and Preliminary Baselines

ARES Research Working Draft

27 October 2026 – Not submission-ready

Abstract

We investigate robot manipulation through reusable feedback skills and learned corrections. This report establishes an executable public-benchmark track and a separate JAKA application track, rather than claiming a new algorithm before its mechanism has been validated. Native ManiSkill stacking experiments demonstrate that a structured feedback controller is already strong: it completes 58 of 64 development-validation cases. An ordinary residual PPO pilot completes 61 cases; forcing the residual to zero recovers the exact feedback outcomes. The learned controller fixes five failures and introduces two. This is preliminary evidence of physical action influence, not evidence of statistical superiority or novelty.

1 Problem and scope

The intended problem is how reusable manipulation skills should adjust legal parameters or recover after observing actual execution outcomes. A local skill may grasp, transport, place or release an object, but reaching an intermediate target does not ensure stable completion. We retain ordinary motion planning and feedback control as strong components. Reinforcement learning is evaluated by physical task success and execution cost, not merely prediction loss.

The public track preserves the default robot, dynamics, reward and success predicate of ManiSkill tasks. It runs in SAPIEN and is not labelled an Isaac Lab experiment. The application track uses JAKA and EG2 assets in Isaac Sim 5.1 and Isaac Lab 2.3.0. Replacing Panda with JAKA produces an adapted task, not an official benchmark score. Neither track demonstrates real-robot deployment.

2 Relation to prior work

STAP studies skill values and planning; RAPS uses parameterized action primitives; ViSkill and Sequential Dexterity study downstream feasibility and chaining; RecoveryChaining combines nominal skills with learned recovery. Residual RL is also established. Consequently, parameterization, skill switching, downstream value and residual addition are not claimed as novel here. The proposed research increment remains a hypothesis: exploit current execution outcomes to select legal skill parameters or bounded recovery actions under a shared budget, with independent causal ablations. A future manuscript must document the exact difference from these methods and cite their original papers after the method is fixed; this document is not a completed literature review.

3 Implemented baseline

Let s_t contain the native simulated robot and object state. A feedback controller has phase z_t and produces $u_0(s_t, z_t)$. Both matched PPO variants observe $o_t = [s_t, \mathrm{onehot}(z_t), u_0(s_t, z_t)]$. The direct baseline executes $u_t = \mathrm{clip}(\pi_\theta(o_t), -1, 1)$; the residual baseline executes

$$u_t = \mathrm{clip}\big(u_0(s_t, z_t) + 0.15\pi_\theta(o_t), -1, 1\big).$$

The native Cartesian delta-position and gripper controller converts this command to physical actuation. No object pose is overwritten during an episode. Phase updates use current geometry and elapsed dwell, not future success labels. PPO, GAE and the upstream network are standard. No new auxiliary head is claimed.

The native state is privileged relative to an unimplemented vision system; it is explicitly a state-based simulation experiment. Controller phase and nominal command are observable internal controller states. The actor does not receive future suffix execution outcomes. The environment reward and 50-step StackCube horizon remain unchanged.

4 Experimental protocol

ManiSkill is pinned to commit `62ff3a5896b4d5b4cf0ac4c8d79afe600c9404a3`. Training uses 64 parallel environments, 50-step rollout windows, eight minibatches and the upstream PPO implementation. Matched Cartesian and residual pilots use 256,000 online interactions each. A separate upstream joint-delta PPO reference uses 1,024,000 interactions. Different action interfaces prevent treating that reference as a fair mechanism ablation.

Development validation uses 16 batches of four environments with batch seeds $27101 + 4k$, $k = 0, \ldots, 15$. Every policy is evaluated on these same cases with deterministic actions. These are development cases, not a sealed final test. Per-step action, clipping, phase, state, reward and native task predicates are retained, together with checkpoint and source hashes. Success is the native final predicate; success-ever is reported separately in raw artifacts.

5 Preliminary results

Controller	Training interactions	Final success
Structured Cartesian feedback	0	58/64
Direct Cartesian PPO, seed 31	256,000	0/64
Ordinary residual PPO, seed 31	256,000	61/64
Ordinary residual PPO, seed 37	256,000	62/64
Ordinary residual PPO, seed 43	256,000	60/64
Same residual checkpoint, forced zero	same checkpoint	58/64
Upstream joint-delta PPO, seed 31	1,024,000	0/64

The residual changes executed Cartesian commands and physical outcomes. It recovers five feedback failures, loses two feedback successes, shares 56 successes and one failure. The exact paired binomial/McNemar test on the seven discordant cases has two-sided $p = 0.453125$. Thus these cases do not establish statistical superiority. The mean norm of the normalized executed-minus-nominal XYZ command is approximately 0.148; it is exactly zero in the zero-residual ablation. Fixed seeds 37 and 43 complete 62 and 60 cases respectively. The three-seed mean is 95.3125%, with sample standard deviation 1.5625 percentage points, versus feedback 90.625%. These repeat the same 64 development cases; they are not 192 independent scenes. Seven final checkpoints and 2,560,000 training interactions were completed across all pilots, excluding separately logged evaluation/scout steps. No final checkpoint was selected by its score.

6 Task diagnostics and application boundary

A GPU planner-observation defect made the native scripted Stack planner use a stale CPU-component OBB instead of the current actor pose. Correcting only this observation adapter changed four paired diagnostic cases from failure to success. However, they required 88–103 steps, beyond the native 50-step budget. They are not official within-budget successes. Corrected PegInsertionSide planning also completed four cases, predominantly beyond its 100-step horizon. These results identify solvability and budget issues, not algorithmic novelty.

Isaac Lab Lift, Stack and Drawer were physically run with one and four environments. Lift and Drawer configurations lack a uniform success termination; measured height and drawer displacement must not be presented as an invented official success rate. JAKA/EG2 drove for 720 physical steps, but independent joint-limit and mimic checks failed. It remains an engineering smoke, not grasp or transfer success. Historical P1/P2 and field evidence remain separate.

7 Next discriminating study and limitations

The immediate study compares ordinary residual control with execution-result- conditioned legal parameter decisions, keeping observation, action constraints, training budget and physical conditions matched. Necessary ablations remove execution-result conditioning and force learned corrections to zero. Extra recovery/suffix interactions count toward the budget. A second contact family and frozen geometry/dynamics splits are required before claiming generalization. More independent training seeds and stronger feedback tuning are required before claiming a repeatable advantage. Current results support baseline viability, not a publishable new method. Asset licensing and JAKA contact qualification also remain open; restricted field data and simulation assets are not included in this public package.